

LAB 2: Transmission, Radio Wave Propagation, and Antenna Characteristics

Objective

- Understand basic mechanisms of radio wave propagation.
- Analyze the relationship between frequency, wavelength, and transmission loss.
- Evaluate antenna parameters such as gain, radiation pattern, and efficiency.

Components & Equipment

- MATLAB – Online or offline version.

Theory

Wireless communication relies on electromagnetic waves propagating through space. The transmitted power spreads out as it travels, leading to attenuation. The received power depends on the distance between transmitter and receiver, frequency of operation, and the environment (free space, urban, rural).

The power received is given by the Friss Equation as

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi d} \right)^2$$

Where: P_r : Received power, P_t : Transmitted power, G_t, G_r : Antenna gains, d : Distance, and λ : Wavelength. Antenna Gain (dBi) represents directional efficiency.

The Free Space Path Loss (FSPL) equation

$$FSPL = \left(\frac{4\pi df}{c} \right)^2 = \left(\frac{4\pi d}{\lambda} \right)^2$$

$$FSPL(dB) = 20\log_{10}(d) + 20\log_{10}(f) + 32.44$$

Practical systems involve many types of obstructions that cause reflections and scattering. Both theoretical and measurement-based models have shown that beyond a certain distance the average received signal power decreases by a factor of $\frac{1}{r^n}$ also seen as logarithmically decreasing with distance according to a $10n \log(r)$ relationship where n is known as the **path loss** exponent. The path loss exponent for different environments is shown in table 1.

Table 1: Path loss exponent for different environments

Environment	Path loss exponent n
Free space	2
Urban area cellular radio	2.5 to 3.5
Shadowed cellular radio	3 to 5
Obstructed in building	4 to 6

Propagation Mechanisms that can affect transmission of signals include;

- Reflection: Occurs when waves hit large surfaces.
- Diffraction: Bending around obstacles.
- Scattering: Caused by small objects or rough surfaces.

Pre-Lab Exercise

1. A transmitter operates at 2.4 GHz. Calculate the free-space path loss at 100 m and 1 km using the Friis equation.
2. If $P_t = 20$ dBm, $G_t = 2$ dBi, $G_r = 0$ dBi, and $PL = 100$ dB, what is P_r in dBm and in mW?

Lab Exercise

Strathmore University plans to deploy a wireless communication network operating at 2.4 GHz. The network must ensure a minimum received power of -75 dBm across the campus.

You are given:

- Transmit power: 20 dBm
- Transmit antenna gain: 2 dBi
- Receive antenna gain: 0 dBi
- Path loss exponent: $n = 3.2$
- Shadowing standard deviation: $\sigma = 6$ dB

Using the log-distance path loss model,

- i. Simulate received power vs distance (10 m to 1000 m) include shadowing (single realization + Monte Carlo)
 - a. Plot Received power without shadowing, Received power with shadowing, and Mean $\pm$ standard deviation
 - b. Determine maximum coverage distance
- ii. Simulate a half-wave dipole radiation pattern and plot the radiation pattern in polar coordinates.
 - a. Introduce antenna misalignment of 0° , 10° , 30° , 60°
 - b. Determine reduction in received power at 100 m and 500 m

- c. Simulate polarization mismatch for 0° , 45° , 90° and explain why signal disappears at 90°
- iii. Using your results, determine:
 - a. Effective coverage radius (realistic case)
 - b. Estimate number of access points required for a campus area of 0.5 km^2
 - c. Suggest Antenna type (omni vs directional) and Placement strategy

Procedure

- i. Open MATLAB (offline version) or log in (online version)
- ii. Open a new MATLAB file (.m file) and name it as *campus_wireless_lab.m*
- iii. Use the provided MATLAB code
- iv. Run the code and observe the following
 - a. Received Power vs Distance
 - b. Shadowing effect
 - c. Mean received power and the Standard deviation (variability)
- v. Record the results in table 2 below

Table 2

Distance (m)	P_r (No Shadowing)	P_r (Shadowing)
100		
500		
1000		

- vi. Identify Minimum Power Requirement given $P_r \geq -75 \text{ dBm}$
- vii. Check MATLAB Command Window *Coverage Distance (ideal): XXX m* and record the readings in table 3

Table 3

Scenario	Coverage Distance (m)
Ideal	
With Shadowing (approx)	

- viii. Run Dipole section of the coded and observe polar plot
- ix. Identify the main lobe direction, null directions, and symmetry and record the observations in table 4

Table 4

Parameter	Observation
Main lobe direction	
Null angles	
Pattern type	

- x. Run Misalignment simulation section and observe plot curves for 0° , 10° , 30° , 60°
- xi. Compare how curves drop with angle. Note: Higher misalignment = lower received power and record the observations in table 5.

Table 5

Distance (m)	0°	10°	30°	60°
100				
500				

- xii. Run Polarization Section and observe plot for 0° , 45° , 90° . Note that at $90^\circ \rightarrow$ signal approaches zero.
- xiii. Record the readings in table 6.

Table 6

Angle	Loss (dB)
0°	
45°	
90°	

- xiv. Observe Combined Effects Plot and determine the total degradation (difference in dB).
- xv. Record your findings in table 7

Table 7

Distance (m)	Ideal P_r (dBm)	Realistic P_r (dBm)
100		
500		

- xvi. Determine the new coverage using realistic curve and find distance where $P_r = -75$ dBm
- xvii. Record your findings in table 8

Table 8

Case	Coverage Radius (m)
Ideal	
Realistic	

- xviii. Given Campus Area of $0.5 \text{ km}^2 = 500,000 \text{ m}^2$, compute the number of Access Points using $N = \frac{\text{Area}}{\pi R^2}$ and record you answer in table 9

Table 9

Parameter	Value
Coverage Radius (R)	
Number of APs	

- xix. Choose between Omni or directional antennas and justify the placement strategy.

MATLAB Code

```

1  %% Strathmore University Wireless Campus Network Design
2  clear; clc; close all;
3
4  %% PARAMETERS
5  Pt_dBm = 20;
6  Gt_dBi = 2;
7  Gr_dBi = 0;
8  freq_Hz = 2.4e9;
9  d = 10:10:1000;
10 n = 3.2;
11 sigma_shadow = 6;
12
13 c = 3e8;
14 lambda = c / freq_Hz;
15
16 %% REFERENCE PATH LOSS
17 d0 = 1;
18 PL_d0 = 20*log10(4*pi*d0 / lambda);
19
20 %% LOG-DISTANCE MODEL
21 PL = PL_d0 + 10*n*log10(d/d0);
22 Pr = Pt_dBm + Gt_dBi + Gr_dBi - PL;
23
24 %% SHADOWING
25 shadow = sigma_shadow * randn(size(d));
26 Pr_shadow = Pr - shadow;
27
28 %% MONTE CARLO
29 N = 500;
30 Pr_mc = zeros(length(d), N);
31
32 for k = 1:N
33     shadow_k = sigma_shadow * randn(size(d));

```

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33     shadow_inst = sigma_shadow * randn(size(d));
34     Pr_many(:,k) = Pt_dBm + Gt_dBi + Gr_dBi - (PL_logdist + shadow_inst);
35 end
36 Pr_mean_dB = mean(Pr_many, 2);
37 Pr_std_dB = std(Pr_many, 0, 2);
38
39 %% Plot results
40 figure;
41 subplot(2,1,1);
42 semilogx(d, Pr_dBm, 'b-', 'LineWidth', 1.5); hold on;
43 semilogx(d, Pr_shadow_dBm, 'r--', 'LineWidth', 1.2);
44 xlabel('Distance (m)'); ylabel('Received Power (dBm)');
45 title('Path Loss Model: n=3.2, \sigma=6 dB');
46 legend('Without Shadowing', 'One Realization with Shadowing');
47 grid on;
48
49 subplot(2,1,2);
50 errorbar(d, Pr_mean_dB, Pr_std_dB, 'ko-');
51 xlabel('Distance (m)'); ylabel('Mean Received Power ± Std (dBm)');
52 title('Monte Carlo Shadowing Statistics (1000 realizations)');
53 grid on; set(gca, 'XScale', 'log');
54
55 %% Compare with Free-space (n=2)
56 PL_freespace = PL_d0 + 10*2*log10(d/d0);
57 Pr_freespace = Pt_dBm + Gt_dBi + Gr_dBi - PL_freespace;
58 figure;
59 semilogx(d, Pr_dBm, 'r-', d, Pr_freespace, 'b--', 'LineWidth', 1.5);
60 xlabel('Distance (m)'); ylabel('Received Power (dBm)');
61 legend('Urban (n=3.2)', 'Free Space (n=2)');
62 title('Effect of Path Loss Exponent on Received Power');
63 grid on;
64
65 %% Antenna gain impact study
66
67 %% Antenna gain impact study
68 Gt_values = [0, 3, 6]; % dBi
69 figure; hold on;
70 for g = 1:length(Gt_values)
71     Pr_gain = Pt_dBm + Gt_values(g) + Gr_dBi - PL_logdist;
72     semilogx(d, Pr_gain, 'LineWidth', 1.2);
73 end
74 xlabel('Distance (m)'); ylabel('Received Power (dBm)');
75 legend('Gt=0 dBi', 'Gt=3 dBi', 'Gt=6 dBi');
76 title('Impact of Transmit Antenna Gain');
77 grid on;

```

Questions

1. Why does misalignment reduce received power?
2. Why is polarization critical in wireless links?
3. Why is realistic coverage smaller than theoretical?
4. What trade-offs exist in antenna selection?

Submission

- The work is to be done in groups of Maximum 5 per group
- The lab report must contain completed tables, necessary calculations, and Network design explanation